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# Plastic film buried in a low-lying strongly salt-affected wasteland: An effective desalinization approach should not be ignored for *Amorpha fruticosa* afforestation

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ABSTRACT

Cutting off the upward movement of salts by capillary barriers may be an effective approach to control the root-zone salt accumulation in a low-lying salt-affected wasteland with high evaporation. This study aimed to determine whether buried plastic film (BPF), as an ignored desalinization approach, could better sustainably ameliorate salt-affected soil environment especially soil salinity than common capillary barriers (buried maize straw, BMS; buried sand, BS) and CK (a control without buried material) during *Amorpha fruticosa* afforestation. All the capillary barriers especially BPF effectively reduced electrical conductivity of saturated paste extracts ($EC_e$) compared to CK in both years mainly due to its greater effectiveness of salts isolation, and gravitational washing increasingly via more preferential pathways indicated by its higher fine-root mass density (FRMD). Capillary barriers varyingly improved the fertility and structure in different soil layers. BPF reduced soil organic carbon (SOC) in 0–20 cm most in 2016, but increased it most in 2017; while BMS increased SOC in 20–40 cm most in both years. In both years, available nitrogen (AN) in 0–20 cm increased most for BPF, while AN in 20–40 cm did for BMS. Similarly, BPF improved better in soil structure (mean weight diameter, MWD; soil total porosity, STP) in 0–20 cm while BMS did in 20–40 cm. As the salinity declined, BPF was more beneficial to the stand establishment and plant growth which in turn further bio-physicochemically improved salt-affected soil environment. BPF was not only far less costly than BMS or BS, but also obviously produced more yield and thus greater net revenue in the second year. Thus, BPF is a promising alternative for desalinization during the afforestation of salt-tolerant plants in the similar strongly salt-affected areas.

## 1. Introduction

Secondary salinization is one of non-ignorable threats causing land degradation in arid and semiarid regions worldwide (Li et al., 2014; Daliakopoulos et al., 2016). Such degradation may cause an irreversible result of desertification and ecosystem collapse, thus seriously threatening the local ecological security. The Yinchuan Plain has prosperous irrigated drylands along the Yellow River of northwestern China, but long-term improper irrigation management raising the groundwater table has contributed to a large increase in surface soil salinity especially of the low-lying drylands due to capillary action under the extremely arid condition of high evaporation (ca., 1825 mm/a) vs. limited precipitation (ca., 185 mm/a) (Zhou et al., 2013). Recently, 33.5% of arable lands are suffering from salinity problem in that region, of which around

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$4.8 \times 10^4$ ha are regarded as strongly salt-affected soils (electrical conductivity of the saturated soil extract, $EC_e$: 8–16 dS/m) abandoned. These soils are universally difficult for conventional farming resulting from their high salinity, low fertility, and poor soil structure (Dong et al., 2020). However, using salt-tolerant plants with deep rooting may sustainably ameliorate the poor structure and function of these salt-contaminated soils for agricultural development in the future (Nikalje et al., 2018).

*Amorpha fruticosa*, as an important leguminous shrub, has been widely planted to revegetate ecologically fragile lands (e.g., salt-affected soil) because of its strong phytoremediation capacity (Han et al., 2016). For example, *A. fruticose* can improve salt-affected soil by the root chemical effects (increasing partial pressure $CO_2$ and $H^+$ release) that help dissolve calcite for the replacement of exchangeable $Na^+$, the root physical effects (creating either biopores or structural cracks for preferential flow; enhancing aggregate stability) that help improve soil structure, and the removal of shoots that uptake salts (e.g., $Na^+$) (Qadir et al., 2007; Cao and Song, 2012; Luo et al., 2019). Although *A. fruticosa* is classified as a moderately salt-tolerance species, its growth could be impaired at $EC_e$ above 6.28 dS/m, causing growth retardation or even establishment failure owing to osmotic effects, ion toxicity ($Na^+$ and $Cl^-$), and nutrient imbalances (Yan et al., 2008). In addition, its early growth is often more vulnerable to soil salinity than other growth stages. Thus, there is a need to effectively alleviate high root-zone salt stress for the successful afforestation in a strongly salt-affected soil.

In the Yinchuan Plain, the temporary reduction of soil salinity solely relying on traditional flood irrigation can re-accumulate in the root zone by capillary action (Zhou et al., 2013; Dong et al., 2020). Over-irrigation with a lack of adequate drainage will raise groundwater table, thus potentially aggravating the salt accumulation (Dong et al., 2009). Although some other leaching and chemical techniques (e.g., drip irrigation, ridge-furrow system, and gypsum application) have been developed to ameliorate strongly salt-affected soils (Wang et al., 2016; Dong et al., 2020), their disadvantages are also obvious such as high expense, intense labor, short-term effect, or potential contamination (Rooney et al., 1998; Li and Wang, 2018; Chávez-García and Siebe, 2019). Similarly, mulching (e.g., plastic film, crop residues, sand or gravel) as a common water-saving approach may reduce water table and suppress salt moving upward to some extent (Dong et al., 2009, 2018). For example, Gao et al. (2010) reported that perforated plastic mulches with different open-hole ratios (0%, 0.69%, 2.78%, 7.71%, 18.7%, and 100%) differentially reduced soil evaporation and topsoil salinity. However, salt accumulation in the root zone under mulching still remains a problem for a low-lying strongly salt-affected soil in the Yinchuan Plain because of improper irrigation management and high evaporation. Overall, these techniques do not have effective capillary cut to restrict the salinity source from the groundwater.

In order to reduce unproductive water loss and restrict salinity source, various buried materials as barriers with the hydraulic characteristics distinctly different from salt-affected soils were implemented to cut off the capillary rise of salts into the root zone (Chen et al., 2016; He et al., 2018). The water in a finer-textured soil with the smaller pores characterized by more negative pressure potential does not readily penetrate the coarser-textured layer; the hydrophobicity of capillary barrier can be another non-negligible resistance especially for that with high organic residue, and lateral flow alongside hydrophobic barrier may also occur (Chávez-García and Siebe, 2019). Thus, capillary barriers can increase root-zone soil moisture and salt leaching efficiency after a flood irrigation event by fully dissolving soluble salts above the barriers (Zhao et al., 2016a). As soil salinity decreases, more preferential pathways are increasingly created by root penetration for further salts leaching. For instance, He et al. (2018) demonstrated that using straw, sand, and fly ash layers could reduce the topsoil salts by 65%, 68.7%, and 60%, respectively. Similarly, Zhao et al. (2016a) buried straw in different depths (20, 40, and 60 cm), of which the 40-cm treatment could best avoid salt emergence into the rooting zone. Shi et al. (2005) also reported that buried sand layer reduced topsoil salinity by at least 70%. Besides, capillary barriers especially from organic residues may also reduce soil pH and increase soil nutrients (He et al., 2018), but their effects on poor soil structure were less assessed (Wang et al., 2012).

To date, however, most common buried materials are associated with straw and sand, less attention has been paid to plastic film that is mainly used as mulching for salts inhibition (Dong et al., 2009; Gao et al., 2010; Wang et al., 2012; Zhao et al., 2016a&b; Dong et al., 2018; Liu et al., 2018) rather than as capillary barrier for salts isolation (Liu et al., 2008; Zhang et al., 2018a), presumably due to the stereotype of the mulching residues' low degradability and permeability in soils (Qin et al., 2021). In addition, previous studies on capillary barriers have mainly focused on crops such as sunflower, tomato, cotton, and maize etc. (Chen et al., 2016; Zhao et al., 2016a&b), or on laboratory experiments (Shi et al., 2005; Chen et al., 2016). Therefore, a two-year afforestation experiment was conducted to explore the effects of capillary barriers, including buried maize straw (BMS), buried sand (BS), and buried plastic film (BPF) with some 2-mm holes, on soil microenvironment and *A. fruticosa* growth in a low-lying strongly salt-affected wasteland. We hypothesized that BPF as a capillary barrier with the distributed micropores of less permeability would better reduce soil salinity mainly due to its greater effectiveness of salts isolation plus gravitational washing, thus facilitating a better performance in the establishment, growth, development, and net revenue.

## 2. Materials and methods

### 2.1. Site description

The field experiment was selected at the Shuxin Forestry Station of the Qingtongxia City, in the Yinchuan Plain of northwestern China (38°1′ N, 105°56′ E, ca. 1139.8 m a.s.l.). The site is traditionally irrigated by the Yellow River through the Plain. It is a typical arid inland climate: mean annual temperature, 9.8 °C; mean monthly temperature ranging from −6.1 °C in January to 23.6 °C in July; mean annual precipitation, only 185 mm; average annual evaporation, 1825 mm (1981–2010; http://data.cma.cn). The groundwater table fluctuates between 0.9 and 2.1 m beneath the ground, with a total dissolved solid (TDS) of up to 15.0 g/L (i.e., $EC_{25} \approx 21.6$ dS/m, Fig. 1). The basic soil properties are summarized in Table 1.

The total precipitation was only 162.0 mm and 140.0 mm during the growth periods in two years (from May to October in 2016–2017), respectively; while the corresponding $ET_0$ was up to 794.4 mm and 931.2 mm (Fig. S1).

### 2.2. Treatment design

The experiment included four treatments at a soil depth of 40 cm: (1) buried maize straw (BMS; 40 cm long × 40 cm wide × 5.00 cm thick); (2) buried sand (BS; 40 cm long × 40 cm wide × 5.00 cm thick); (3) buried plastic film (BPF; 40 cm long × 40 cm wide × 0.08 mm thick) with 2-mm holes (open-hole ratio: ca., 0.51%); and (4) a control with no capillary barrier (CK) (Fig. S2). The experiment had a randomized complete block design with three replications (more details see Fig. S2). Except for installing buried materials, other field management (tillage, drainage ditch, and flood irrigation, etc.) was the same as in our previous study (Liu et al., 2018).

### 2.3. Data collection

#### 2.3.1. Soil salinity

Soil samples from each plot were collected in June, August, and October during three growth periods (May–June, July–August, and September–October for the early/middle/later growth periods, respectively) in both 2016 and 2017. Soil samples in each same depth (0–20, 20–40, 40–60 cm) were collected at 20 cm away from three

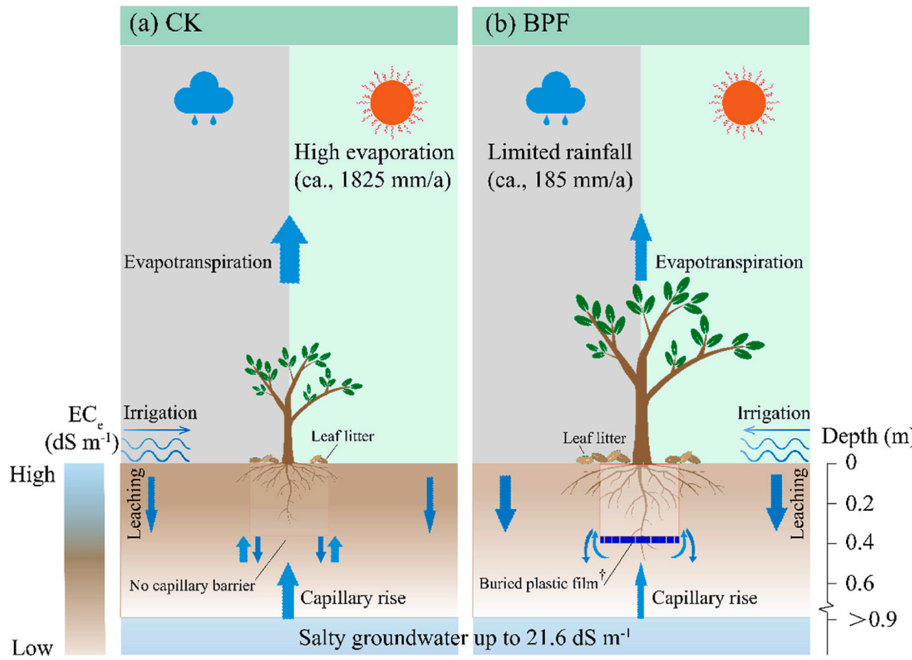


**Fig. 1.** The schematic diagram of (a) CK (control without capillary barrier) with more salts re-accumulation even after flood irrigation and (b) BPF (buried plastic mulch with 2-mm holes that allow fine roots penetration) with the better reduction of the root-zone salinity mainly due to its greater effectiveness of salts isolation, and gravitational washing increasingly via a series of bio-physicochemical actions.

**Table 1**
The basic soil physicochemical properties.

| Depths | Soil particle contents (%) | | | Soil | $EC_e$ | $pH_e$ | $SAR_e$ | SOM | TN (g $kg^{-1}$) | STP |
|---|---|---|---|---|---|---|---|---|---|---|
| (cm) | Clay | Silt | Sand | texture | (dS $m^{-1}$) | | $(mmol_c/L)^{0.5}$ | (g $kg^{-1}$) | | (%) |
| 0–20 | 2.23 | 58.48 | 39.29 | Silt loam | 8.77 | 8.50 | 50.63 | 14.7 | 0.39 | 35.85 |
| 20–40 | 1.46 | 57.36 | 41.18 | Silt loam | 8.58 | 8.53 | 51.21 | 13.8 | 0.26 | 36.23 |
| 40–60 | 1.55 | 53.81 | 44.65 | Silt loam | 8.48 | 8.39 | 41.33 | 6.5 | 0.17 | 35.85 |
| 60–80 | 1.84 | 58.92 | 39.25 | Silt loam | 8.40 | 8.43 | 36.84 | 7.9 | 0.16 | 36.60 |

Note: $EC_e$, $pH_e$, and $SAR_e$, refers to electrical conductivity, pH, and sodium adsorption ratio of a saturated paste extract, respectively; SOM, soil organic matter; TN, total nitrogen; STP, soil total porosity.

representative seedlings, and then put together to form a composite sample. The soil samples were air-dried, ground, and passed (via a 1-mm sieve) to determine soil salinity ($EC_e$, dS $m^{-1}$) (Supplementary note 1).

### *2.3.2. Soil moisture and nutrients*

The fresh subsamples were also used to measure soil water content (SWC, %) by the gravimetric method. Like $EC_e$, the mean SWC was also calculated for each soil layer.

Soil samples were obtained in the same layers with three replicates in October of 2016 and 2017. They were used to analyze soil organic carbon (SOC, g $kg^{-1}$) and available nitrogen (AN, including $NO_3$–N and $NH_4$–N; mg $kg^{-1}$) by the dichromate oxidation method, and the salicylate-nitroprusside method via a Discrete Auto Analyzer (Smartchem200, AMS/Westco, Italy) (Singh et al., 2013), respectively.

### *2.3.3. Soil structure*

The soil subsamples, collected in October 2017, were used to measure aggregate stability. Their large clods were gently broken into smaller aggregates before air-dried and passed through 10 mm sieve by the wet sieving method (Supplementary note 2; Luo et al., 2018). The aggregate stability was calculated by mean weight diameter (MWD, mm):

$$MWD = \sum_{i=1}^{n} X_i \times W_i \tag{1}$$

where $X_i$ indicates the mean diameter for each aggregate size class $i$ (>5, 2–5, 1–2, 0.5–1, 0.25–0.5, and <0.25 mm); $W_i$ represents the mass proportion of the corresponding aggregate size class (%); and $n$ is the total number of sieves.

Meanwhile, soil total porosity (STP, %) was calculated by:

$$STP = \left(1 - \frac{SBD}{2.65}\right) \times 100\% \tag{2}$$

where SBD means soil bulk density (g $cm^{-3}$); and 2.65 (g $cm^{-3}$) is the particle density.

### *2.3.4. Stand establishment, growth and yield*

The numbers of surviving seedlings were counted for stand establishment (SE, %; Liu et al., 2018) at three different growth stages in the first year. Some other important growth indicators, including height (H, cm), ground diameter (GD, cm), crown diameter (CD, cm), and leaf number (LN) and area (LA, $cm^2$), were investigated at three growth stages.

Root samples (0–20, 20–40, 40–60 cm) were collected with three replicates for each treatment in October of 2016 and 2017 (Supplementary note 3). Fine-root traits, including fine-root mass density (FRMD, mg $cm^{-3}$; soil volume, 40 × 40 × 20 $cm^3$) and specific root length (SRL, cm $g^{-1}$), were calculated as shown in Supplementary note 3.

At the later growth stage of each year, *A. fruticosa* individuals were manually harvested and the corresponding seed yields (kg ha$^{-1}$) were determined after traditional sun-drying.

#### 2.3.5. Cost recording

Material input included seedling, pesticide, irrigation, and buried materials. Labor input here was defined as the prices of rototiller operator for soil tillage, excavator operator for drainage ditch, and other field assistants. Input value was summed according to the prices of the material and labor, while output value did based on mean yearly market price. Both the input and output values (CNY, ¥) were converted to dollars ($) with the official exchange rate.

### 2.4. Statistical analysis

First of all, both normality and homoscedasticity of variance should be checked via Shapiro-Wilk test and Levene test, respectively. Then, one-way ANOVA was used to evaluate the effect of treatments on soil properties and plant growth parameters. Statistical difference between a pair of means was found by the significant ANOVA via the Fisher's LSD test at $P < 0.05$. A three-way ANOVA test was used to identify the interaction effects of the treatment type, soil depth and sampling year on soil properties and fine root traits. Both Pearson's correlation analysis and principal component analysis (PCA) were conducted to find out their associations. IBM SPSS ver. 22.0 program (IBM Corp., Armonk, NY, USA) was used for data analysis, while OriginPro 2021 ver. 9.8.0.200 software (OriginLab Corp., Northampton, MA, USA), Adobe Illustrator 2020 ver. 24.0.1 (see Fig. 1) (Adobe Systems Inc., San Jose, CA, USA) and AutoCAD 2010 ver. R18 (Fig. S2) (Autodesk Inc., San Rafael, CA, USA) for graphics drawing.

## 3. Results

### 3.1. The reduction of soil salinity as affected by capillary barriers

All the capillary barriers, especially BPF, effectively reduced soil salinity compared to CK in both years ($P < 0.05$, Table 2). In 2016, $EC_e$ to a depth of 60 cm for BMS, BS, and BPF reduced by 23.64, 15.31, and 31.59%, respectively. The reductions in $EC_e$ for BMS, BS, and BPF also occurred in 2017, with the decreases by 32.23, 16.43, and 34.18%, respectively.

### 3.2. Effects of capillary barriers on soil moisture and nutrients

SWC retained in some depths of capillary barriers, although generally there were no differences ($P > 0.05$) between capillary barriers about half a month after irrigation (Table 2). In both years BPF always significantly increased more SWC in 20–40 cm (by 16.18% and 10.94% in 2016 and 2017, respectively) and 40–60 cm (by 26.81% and 15.02%) ($P < 0.05$); while BMS only increased it in 40–60 cm (by 23.07%) in 2016 and BS did in 20–40 cm (by 11.61%) in 2017 ($P < 0.05$). As expected, however, there was no difference in SWC in the topsoil (0–20 cm) between any of the capillary barriers and CK ($P > 0.05$).

In 2016, the SOC in the topsoil significantly decreased by 17.28, 9.33, and 21.26%, respectively, for the capillary barriers ($P < 0.05$), in which no difference was found between BMS and BPF ($P > 0.05$); whereas in 20–40 cm only that for BMS increased by 26.70% ($P < 0.05$). In 2017, the SOC in the topsoil increased only for BPF (by 13.82%) ($P < 0.05$); the SOC in 20–40 cm for BS and BPF (apart from BMS with a continuing increase of 35.77%), became much higher (by the increases of 17.04 and 27.72%, respectively) than that for CK ($P < 0.05$) (Table 2).

In both years, the AN in 0–20 cm increased most for BPF (26.80% in 2016; 37.56% in 2017), followed by BMS (13.42, 23.11%) and BS (11.50, 11.57%) ($P < 0.05$). For AN in 20–40 cm, only BMS increased it by 110.79% in 2016, while in 2017 all the capillary barriers did by 58.02 (BMS), 19.38 (BS), and 31.25% (BPF) ($P < 0.05$).

### 3.3. The differential improvement of soil structure

#### 3.3.1. Soil aggregate

After 2 years amelioration, the size distribution of aggregates changed obviously mainly in 0–40 cm especially for BMS and BPF (Fig. S3). In the topsoil, BPF significantly increased large macro-aggregates (>2 mm) and some small macro-aggregates (0.5–2 mm), and reduced micro-aggregate (<2 mm) ($P < 0.05$). In 20–40 cm, however, BMS had higher proportions of large (>2 mm) and some small (1–2 mm) macro-aggregates than did BPF ($P > 0.05$). Overall, the MWD

**Table 2**
Dynamics of soil properties for different capillary barriers in 2016–2017.

| Treatments | Soil properties | Soil depths (cm) in 2016 0–20 | 20–40 | 40–60 | | Soil depths (cm) in 2017 0–20 | 20–40 | 40–60 |
|---|---|---|---|---|---|---|---|---|
| BMS | $EC_e$ | 6.14 ± 0.24 b | 5.87 ± 0.39 b | 6.36 ± 0.74 b | | 5.23 ± 0.55 c | 5.14 ± 0.22 c | 4.78 ± 0.31 b |
| BS | (dS m$^{-1}$) | 6.73 ± 1.24 b | 7.39 ± 1.39 a | 6.26 ± 0.86 b | | 6.24 ± 0.29 b | 6.18 ± 0.24 b | 6.22 ± 0.86 a |
| BPF | | 5.87 ± 1.13 b | 5.52 ± 0.23 b | 5.17 ± 0.11 c | | 4.81 ± 0.42 c | 5.01 ± 0.41 c | 4.86 ± 0.36 b |
| CK | | 9.81 ± 1.66 a | 7.47 ± 0.52 a | 7.24 ± 0.71 a | | 7.83 ± 1.02 a | 7.78 ± 1.46 a | 6.79 ± 0.50 a |
| *F* values | | T | SD | Y | T × SD | T × Y | SD × Y | T × SD × Y |
| | | 27.99*** | 3.99* | 222.74*** | 0.58 ns | 2.88* | 0.96 ns | 0.79 ns |
| BMS | SWC | 17.88 ± 1.67 a | 20.60 ± 2.23 ab | 21.11 ± 2.64 a | | 14.84 ± 2.69 a | 17.88 ± 2.33 ab | 19.00 ± 1.43 ab |
| BS | (%) | 17.53 ± 2.64 a | 19.85 ± 2.45 ab | 19.47 ± 2.96 ab | | 15.65 ± 2.05 a | 18.32 ± 1.51 a | 18.04 ± 1.56 b |
| BPF | | 18.65 ± 1.86 a | 21.69 ± 3.42 a | 21.75 ± 2.35 a | | 15.47 ± 2.12 a | 18.21 ± 1.66 a | 19.94 ± 1.72 a |
| CK | | 17.15 ± 1.87 a | 18.67 ± 4.08 b | 18.04 ± 2.68 b | | 14.54 ± 2.42 a | 16.41 ± 1.70 b | 17.33 ± 1.17 b |
| *F* values | | T | SD | Y | T × SD | T × Y | SD × Y | T × SD × Y |
| | | 3.77* | 25.96** | 90.43*** | 0.62 ns | 0.45 ns | 0.30 ns | 0.47 ns |
| BMS | SOC | 6.03 ± 0.17 c | 7.07 ± 0.40 a | 2.43 ± 0.19 a | | 7.22 ± 0.10 b | 7.25 ± 0.24 a | 2.62 ± 0.12 a |
| BS | (g kg$^{-1}$) | 6.61 ± 0.4 b | 5.34 ± 0.31 b | 2.35 ± 0.07 a | | 7.19 ± 0.06 b | 6.25 ± 0.11 c | 2.44 ± 0.15 ab |
| BPF | | 5.74 ± 0.23 c | 5.25 ± 0.10 b | 2.28 ± 0.17 a | | 7.99 ± 0.13 a | 6.82 ± 0.33 b | 2.37 ± 0.09 b |
| CK | | 7.29 ± 0.39 a | 5.58 ± 0.13 b | 2.18 ± 0.10 a | | 7.02 ± 0.15 b | 5.34 ± 0.02 d | 2.32 ± 0.10 b |
| *F* values | | T | SD | Y | T × SD | T × Y | SD × Y | T × SD × Y |
| | | 19.89*** | 3360.25*** | 133.45*** | 33.66*** | 37.12*** | 24.04*** | 13.29*** |
| BMS | AN | 25.28 ± 1.90 b | 29.78 ± 2.17 a | 6.13 ± 0.33 a | | 22.32 ± 1.08 ab | 25.03 ± 2.27 a | 8.74 ± 0.69 a |
| BS | (mg kg$^{-1}$) | 24.85 ± 0.08 b | 14.72 ± 2.56 b | 5.93 ± 0.06 a | | 20.98 ± 1.98 b | 18.91 ± 1.46 b | 7.72 ± 1.48 a |
| BPF | | 28.26 ± 0.38 a | 14.77 ± 1.58 b | 5.75 ± 0.47 a | | 24.94 ± 1.42 a | 20.79 ± 1.49 b | 7.78 ± 1.20 a |
| CK | | 22.29 ± 0.37 c | 14.13 ± 0.71 b | 5.07 ± 0.09 a | | 18.13 ± 2.09 c | 15.84 ± 0.65 c | 7.19 ± 0.49 a |
| *F* values | | T | SD | Y | T × SD | T × Y | SD × Y | T × SD × Y |
| | | 53.52*** | 881.05*** | 0.46 ns | 29.91*** | 3.76* | 28.15*** | 5.70*** |

Note: Different lowercase letters in the same column indicates a significant difference between treatments ($P < 0.05$). *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ns, not significant.

in the topsoil increased most for BPF (74.30%), followed by BMS (50.79%) and BS (34.95%); while in 20–40 cm that did most for BMS (106.60%), followed by BPF (65.65%) and BS (38.40%) (Fig. 2a).

#### 3.3.2. Soil total porosity

Similarly, the significant improvements in STP as affected by different capillary barriers were mainly distributed in 0–40 cm (Fig. 2b). Specifically, in 0–20 cm BPF performed best in improving its STP (increase of 10.24%) in the second year, followed by BMS (5.41%), and BS (4.66%). In 20–40 cm, however, BMS improved better in STP (20.44%), followed by BPF (16.47%), and BS (14.20%).

### 3.4. Effects of capillary barriers on Amorpha fruticosa afforestation

#### 3.4.1. Aboveground growth

The SE varied with transplanting time for capillary barriers (Fig. 3). At the early growth stage (May–June) of the first year, CK had a higher (although non-significant, $P > 0.05$) SE than that of the capillary barriers. Thereafter, however, the SE began to be greater for those capillary barriers, in which only BPF had significant increases both by 20.74% at the middle and later stages ($P < 0.05$).

For other growth indicators (H, GD, CD, LN, and LA), at the first early growth stage in 2016, no significant difference was detected in the whole growth indicators between treatments (Table 3). After the first middle growth stage, those indicators became generally highest for BPF, while no remarkable significance existed between CK and BMS or BS until the second early growth stage in 2017.

#### 3.4.2. Root traits as proxies for preferential flow

Obviously, BPF, followed by BMS and then BS, had a highest FRMD to a depth of 40 cm in both years (Fig. 4a&b). Surprisingly, apart from BMS and BS, BPF also had fine roots that reached the 40–60 cm layer in the second year (Fig. 4b). In contrast to FRMD, SRL values were always highest in CK at 0–40 cm (Fig. 4c).

### 3.5. Relationships between soil physicochemical properties and plant indicators

$EC_e$ was significantly negatively correlated with most growth indicators (except for SRL) (Fig. 5a). SWC had a significant negative relationship with CD, LN, FRMD, and yield, respectively (Fig. 5a). Both SOC and AN were significantly positively correlated with FRMD, but negatively with SRL. MWD was positively correlated with SE, GD, and LA. CK tended to span a narrow space at one side associated with higher soil salinity and worse plant growth, whereas BPF tended to occupy the opposite side of the spectrum exhibiting lower soil salinity and better growth performance (Fig. 5b, Table S1).

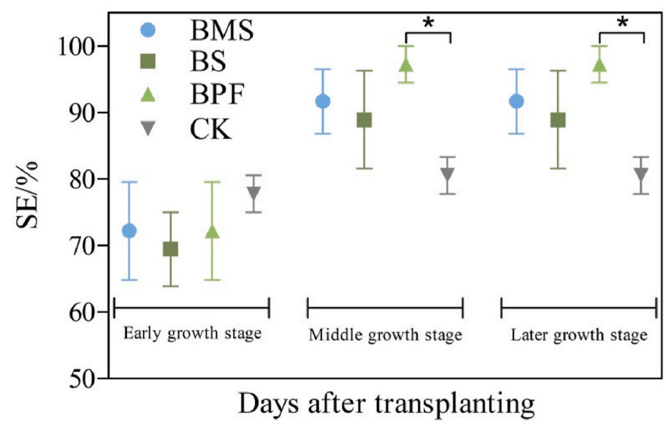


**Fig. 3.** The SE (stand establishment) of *Amorpha fruticosa* as affected by different capillary barriers in 2016.

### 3.6. Input, output, and net revenue

In 2016, the capillary barriers required additional buried materials and labors for installing barrier layer, in which the cost of buried material was lowest for BPF (6.28 $/ha), but highest for BS (188.76 $/ha), followed by BMS (41.77 $/ha). In 2017, however, capillary barriers especially BPF had obviously higher net revenues than did CK, mainly because of their increased seed yields. Summed across 2 years, BPF earned the net revenue of 650.56$ per hectare, while BMS and BS still had negative net benefits.

## 4. Discussion

Up to now, most studies of capillary barriers have focused on common materials such as straw (e.g., maize, rice) and sand (Wang et al., 2012; Zhao et al., 2016a&b; He et al., 2018), but far too little attention has been paid to plastic film for salts isolation probably traditionally regarded as a mulching material of low degradability (Qin et al., 2021). Instead, the single-use amount of plastic film as capillary barrier can be negligible in contrast to that of agricultural application in each year (2.92 kg/ha once for all in BPF vs. 45 kg/ha per year in agriculture according to Dong et al., 2009). What's more, the characteristic that plastic film does not readily degrade in soil makes it as a viable

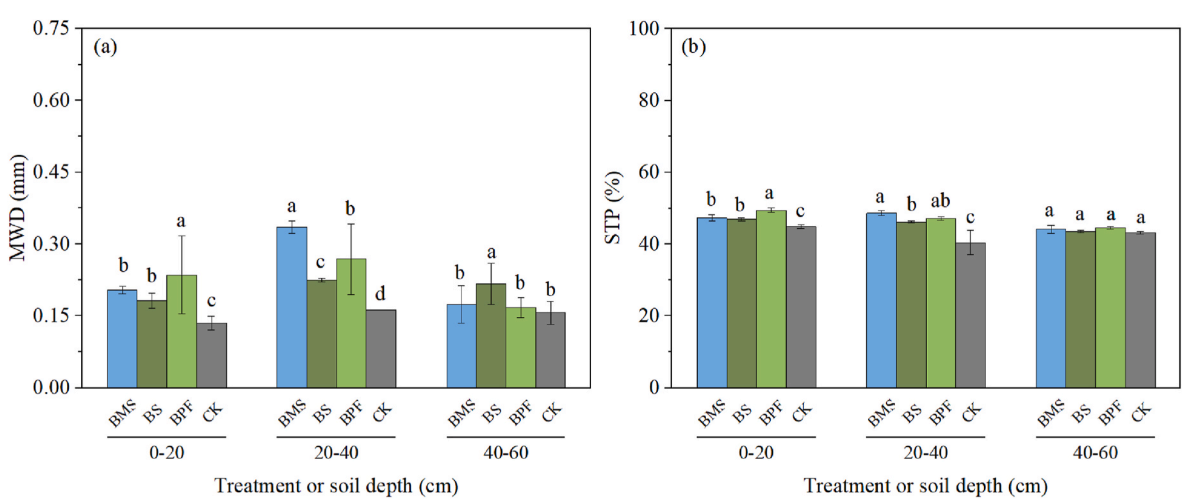


**Fig. 2.** (a) The MWD (mean weight diameter) of soil aggregation and (b) STP (soil total porosity) as affected by different capillary barriers in 2017.

**Table 3**
The aboveground growth of *Amorpha fruticosa* as affected by different capillary barriers in two years.

| Treatments | H/cm | GD/cm | CD/cm | LN/no. | LA/cm$^2$ |
|---|---|---|---|---|---|
| *The early growth stage in 2016* | | | | | |
| BMS | 75.9 ± 19.3 a | 0.98 ± 0.28 a | 18.1 ± 7.0 a | 551.0 ± 222.4 a | 1.20 ± 0.12 a |
| BS | 72.2 ± 22.4 a | 0.98 ± 0.13 a | 18.8 ± 6.7 a | 504.9 ± 131.2 a | 1.21 ± 0.14 a |
| BPF | 78.3 ± 15.8 a | 1.03 ± 0.17 a | 19.9 ± 6.2 a | 565.3 ± 158.1 a | 1.31 ± 0.20 a |
| CK | 69.1 ± 23.0 a | 0.97 ± 0.17 a | 17.8 ± 5.8 a | 501.5 ± 120.8 a | 1.18 ± 0.15 a |
| *The middle growth stage in 2016* | | | | | |
| BMS | 96.3 ± 27.0 b | 1.22 ± 0.23 b | 48.2 ± 10.0 b | 2337.0 ± 808.3 b | 1.36 ± 0.29 b |
| BS | 99.5 ± 24.7 b | 1.28 ± 0.26 b | 50.3 ± 14.8 b | 2263.5 ± 888.8 b | 1.37 ± 0.54 b |
| BPF | 112.2 ± 15.5 a | 1.52 ± 0.25 a | 73.5 ± 14.9 a | 4720.6 ± 1824.7 a | 1.66 ± 0.51 a |
| CK | 93.7 ± 21.2 b | 1.21 ± 0.16 b | 54.2 ± 9.6 b | 2562.0 ± 1306.2 b | 1.32 ± 0.21 b |
| *The later growth stage in 2016* | | | | | |
| BMS | 98.5 ± 9.6 b | 1.63 ± 0.32 b | 53.2 ± 10.0 b | 1836.3 ± 516.1 b | 1.34 ± 0.24 b |
| BS | 105.6 ± 10.6 b | 1.63 ± 0.18 b | 51.7 ± 11.2 b | 1962.9 ± 612.4 b | 1.23 ± 0.25 b |
| BPF | 137.1 ± 10.8 a | 2.14 ± 0.28 a | 88.5 ± 14.0 a | 4754.7 ± 1197.9 a | 3.10 ± 1.10 a |
| CK | 95.8 ± 15.1 b | 1.57 ± 0.21 b | 56.3 ± 15.9 b | 1389.8 ± 986.2 b | 1.19 ± 0.46 b |
| *The early growth stage in 2017* | | | | | |
| BMS | 107.6 ± 10.4 b | 1.64 ± 0.23 b | 59.4 ± 13.2 b | 3866.5 ± 327.3 a | 2.36 ± 0.52 b |
| BS | 104.7 ± 9.6 b | 1.63 ± 0.20 b | 55.1 ± 10.7 b | 3133.3 ± 1199.6 a | 2.10 ± 0.73 b |
| BPF | 141.1 ± 11.1 a | 2.17 ± 0.40 a | 97.8 ± 26.5 a | 3685.4 ± 778.2 a | 3.23 ± 0.58 a |
| CK | 94.1 ± 8.8 c | 1.58 ± 0.06 b | 57.2 ± 11.2 b | 2679.1 ± 1726.5 a | 1.66 ± 0.30 c |
| *The middle growth stage in 2017* | | | | | |
| BMS | 151.0 ± 17.4 b | 2.71 ± 0.35 b | 126.1 ± 16.3 b | 4874.0 ± 991.8 ab | 3.31 ± 0.58 a |
| BS | 132.0 ± 20.3 b | 2.17 ± 0.26 c | 98.2 ± 30.2 c | 3233.8 ± 1063.3 b | 3.07 ± 0.58 a |
| BPF | 180.9 ± 29.6 a | 3.44 ± 0.63 a | 170.0 ± 36.5 a | 6497.4 ± 1052.2 a | 3.60 ± 0.43 a |
| CK | 105.4 ± 18.7 c | 1.93 ± 0.46 c | 80.4 ± 27.5 c | 3068.7 ± 2101.7 b | 1.51 ± 0.28 b |
| *The later growth stage in 2017* | | | | | |
| BMS | 152.0 ± 8.0 b | 2.78 ± 0.34 b | 133.0 ± 17.0 b | 5973.0 ± 2277.2 a | 2.81 ± 0.54 ab |
| BS | 141.2 ± 15.4 b | 2.32 ± 0.14 c | 120.2 ± 26.8 bc | 3656.0 ± 1964.9 b | 2.13 ± 0.41 bc |
| BPF | 182.8 ± 27.4 a | 3.55 ± 0.32 a | 178.7 ± 29.6 a | 7070.0 ± 1415.0 a | 3.61 ± 0.94 a |
| CK | 108.6 ± 6.4 c | 1.99 ± 0.20 c | 90.6 ± 24.8 c | 1347.3 ± 417.1 c | 1.39 ± 0.30 c |

Note: H means height; GD, ground diameter; CD, crown diameter; LN, leaf number; LA, leaf area.

advantage to constantly isolate the upward migration of salts in a strongly salt-affected lowland. Some positive evidences also show that BPF is a promising alternative to sustainably ameliorate the strongly salt-affected wasteland.

### 4.1. The differentially improved soil environments by capillary barriers with BPF being superior in reducing soil salinity

Like previous studies (e.g., Zhao et al., 2016a&b), the capillary barriers in this experiment also inhibited the re-accumulation of salts in the region with high evaporation. We found that BPF was more effective than other treatments in reducing soil salts in both years (Table 2). No significant difference occurred in salts reduction (0–40 cm) between BMS and BS, which is consistent with that of He et al. (2018). It might be that, apart from isolating more salts, plastic film (with holes) as the least permeable material could more fully dissolve soil salts by extending soil water residence time before gravitational leaching through and/or alongside the barrier (lateral flow) in an irrigation event (Chávez-García and Siebe, 2019). This could be supported by the highest SWC in BPF (Table 2), where high SWC would facilitate more ion exchange, and reduce more dissolved salts (Zhao et al., 2016a). Meanwhile, BPF with greater root biomass (Fig. 4) may have provided more $Ca^{2+}$ to exchange with $Na^+$ by dissolving calcite through acidic exudates release (eg., $CO_2$ and proton [$H^+$]) (Qadir et al., 2000), and more preferential pathways by root penetration (Luo et al., 2019), thus leaching more salts after irrigation. The salts accumulated in plant organs with greater biomass for BPF may also partly contribute to reducing more soil salinity (Han et al., 2015).

Although irrigation is beneficial to alleviate the root-zone salt stress, high SWC may be a double-edged sword. For example, all the capillary barriers stored more SWC that contributed to leaching more soil salts after flood irrigation compared to CK (Table 2), which, to some extent, could retard especially their early growth and yields (Figs. 3 and 5b) associated with the reduced soil oxygen and nutrient availability (Fig. 5a; Orchard et al., 2016). Nevertheless, it seems that the positive effects of soil salinity reduction on yields by the capillary barriers may have partly counteracted the temporarily negative effects of more SWC.

Both SOC and AN as key components of soil fertility varied with treatments. In 2016, SOC in 0–20 cm significantly decreased for all the capillary barriers especially for BPF, but SOC in 20–40 cm only increased for BMS (Table 2) which agrees with the result of Huo et al. (2017). That reduction in SOC in 0–20 cm may be ascribed to the increasing uptake of nitrogen by both roots and soil microbes as the salinity decreased (Fig. 5a), thus causing enhanced SOC decomposition by rhizosphere priming effect and microbial mining for N (Dijkstra et al., 2021). The greatest reduction of SOC in 0–20 cm for BPF suggests that more nutrients would be absorbed by its individuals. In the second year, conversely, SOC in 0–20 cm increased only for BPF, indicating that its soil fertility was gradually improved. This finding may be caused by the increases in litter decomposition and root exudates (Terrer et al., 2021). Meanwhile, SOC in 20–40 cm for BPF and BS began to be higher than CK, implying their SOC gain exceeded the corresponding loss. Although the root decomposition and root exudates could contribute to SOC gain, BPF was still inferior to BMS with extra organic straw in improving SOC in 20–40 cm. In both years, the AN in the topsoil increased most for BPF, likely due to the differentially increased soil mineralization driven by reduced soil salinity and then enhanced microbial activity (Huo et al., 2017). In 2016, unlike BMS, BPF and BS did not increase in AN in 20–40 cm, while in 2017 BPF still had lower increase in AN than BMS with the extra source of nitrogen. It's worth noting that the nitrogen loss from ammonia volatilization (Chávez-García and Siebe, 2019) and AN (e.g., $NO_3$–N) leaching (Zhang et al., 2018b) may also have occurred especially for BPF (Table 2), further worsening its nutrient stress especially in the early recovering stage. Thus, there is a need to explore the rational application of absorbable nutrients (e.g., seaweed fertilizer) (Yuan et al., 2019) to improve plant's salt tolerance in that stage in future studies.

### 4.2. Capillary barriers varyingly improving the poor structure of different soil layers

In previous studies, more attention has been paid to the influence of incorporation and/or mulching (Liu et al., 2018; Luo et al., 2019; Xie et al., 2020), rather than capillary barrier, in terms of improving poor soil structure, while that of capillary barrier mainly focused on salinity reduction and nutrients improvement (Zhao et al., 2016a&b; Huo et al., 2017). Both BPF and BMS significantly increased soil aggregation formation at 0–40 cm layer, in which BPF better improved in MWD in 0–20 cm while BMS did in 20–40 cm (Fig. 2). However, Danga and Wakindiki (2009) found no difference between BMS and CK in a non-salt-affected

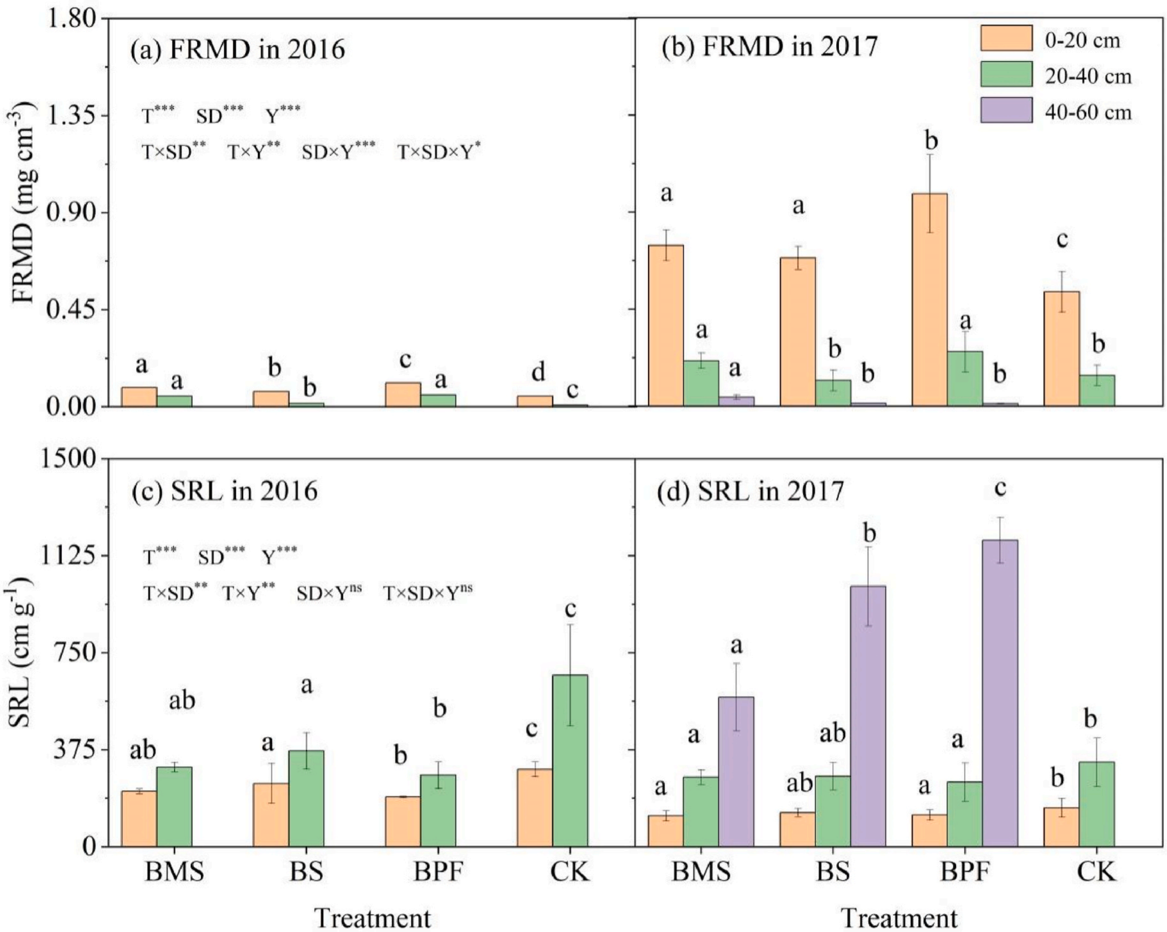


**Fig. 4.** Fine-root characteristics (FRMD, fine root mass density; SRL, specific root length) of *Amorpha fruticosa* for different capillary barriers in two years.

soil. Some reasons may explain the increase of their soil aggregates. On the one hand, the salinity reduction can alleviate the dispersion of soil particles (0.5–1, 1–2, 2–5, and >5 mm, Fig. S3), which was supported by the significant negative correlation between MWD and $EC_e$ (Fig. 5a). Excessive $Na^+$ will disperse soil particles by replacing $Ca^{2+}$ and $Mg^{2+}$, thus causing the breakup of soil aggregates (Wong et al., 2010), whereas salinity reduction can protect soil aggregates from dispersion (Xie et al., 2020). On the other hand, that reduction could also improve soil aggregate by increasing soil biota (i.e., microorganisms, fauna, and roots) and thus their cementing substances (e.g., secretions and mucilages) (Cardoso et al., 2013). Meanwhile, roots and their associated fungi physically enmesh soil into aggregates (Luo et al., 2019). Thus, the greater MWD in 0–20 cm for BPF may result from its more reduction in $EC_e$ (Table 2) and more roots (Fig. 4) that suggest more aggregation formation of soil particles. In addition, the more leaf litter of BPF (Fig. 1 &Table 3) not only provided more organic matter to bind mineral particles into aggregates (Luo et al., 2019), but also protected its soil aggregates from the erosion of wind and rainfall (Prosdocimi et al., 2016). In contrast, the greater MWD in 20–40 cm for BMS could mainly be attributed to its additional organic input which boosted saline soil aggregation formation ($P < 0.05$; Fig. 5).

Similarly, BPF performed best in improving STP in 0–20 cm while BMS did in 20–40 cm. As their soil salinity decreased (Fig. 5a), BPF and BMS may have provided the better soil environment (Table 2) for the growth of roots (Fig. 4a and b) and soil microbes (Singh et al., 2013) in 0–20 and 20–40 cm, respectively. Collectively, those bio-physicochemical factors, as mentioned earlier, were beneficial to form soil aggregates to improve soil structure, thus varyingly increasing their corresponding STP.

### 4.3. Plant establishment, growth, and yield

Good stand establishment and vigorous seedling growth could also reflect the improvement of belowground salt-affected soils (Dong et al., 2009). In our study, however, CK had a slightly greater SE than other capillary barriers in the early recovering stage of the first year (Fig. 3), owing to the capillary barriers' more dissolved salts before leaching that caused more pressure on their early root production. After that stage, the capillary barriers came to have higher SE, of which BPF basically performed better (Fig. 3), likely due to its more suitable soil environment especially the better control of root-zone salinity (Fig. 5a). Similarly, BPF also had better performance in the other plant growth indicators (Table 3). In contrast, the seedlings in other treatments especially CK had to allocate more energy to produce roots when exposed to high soil salinity that imposes osmotic stress, specific-ion effects, and nutritional imbalances on its seedling growth (Dong et al., 2009).

Root traits can be used as proxies for preferential flow. In our study, FRMD to a depth of 40 cm was significantly higher for BPF in both years (Fig. 4a&b), further indicating that *A. fruticosa* created more preferential pathways for salinity leaching (Luo et al., 2019). However, the SRL values at depths of 0–40 cm were always higher for CK (Fig. 4c&d), reflecting its less preferential flow created by root penetration at 0–40 cm which was unfavorable for salinity reduction.

In the first year, BPF was less costly than both BMS and BS, because of its less additional inputs for the buried material. Although the inputs of capillary barriers were far greater than the outputs in the first year,

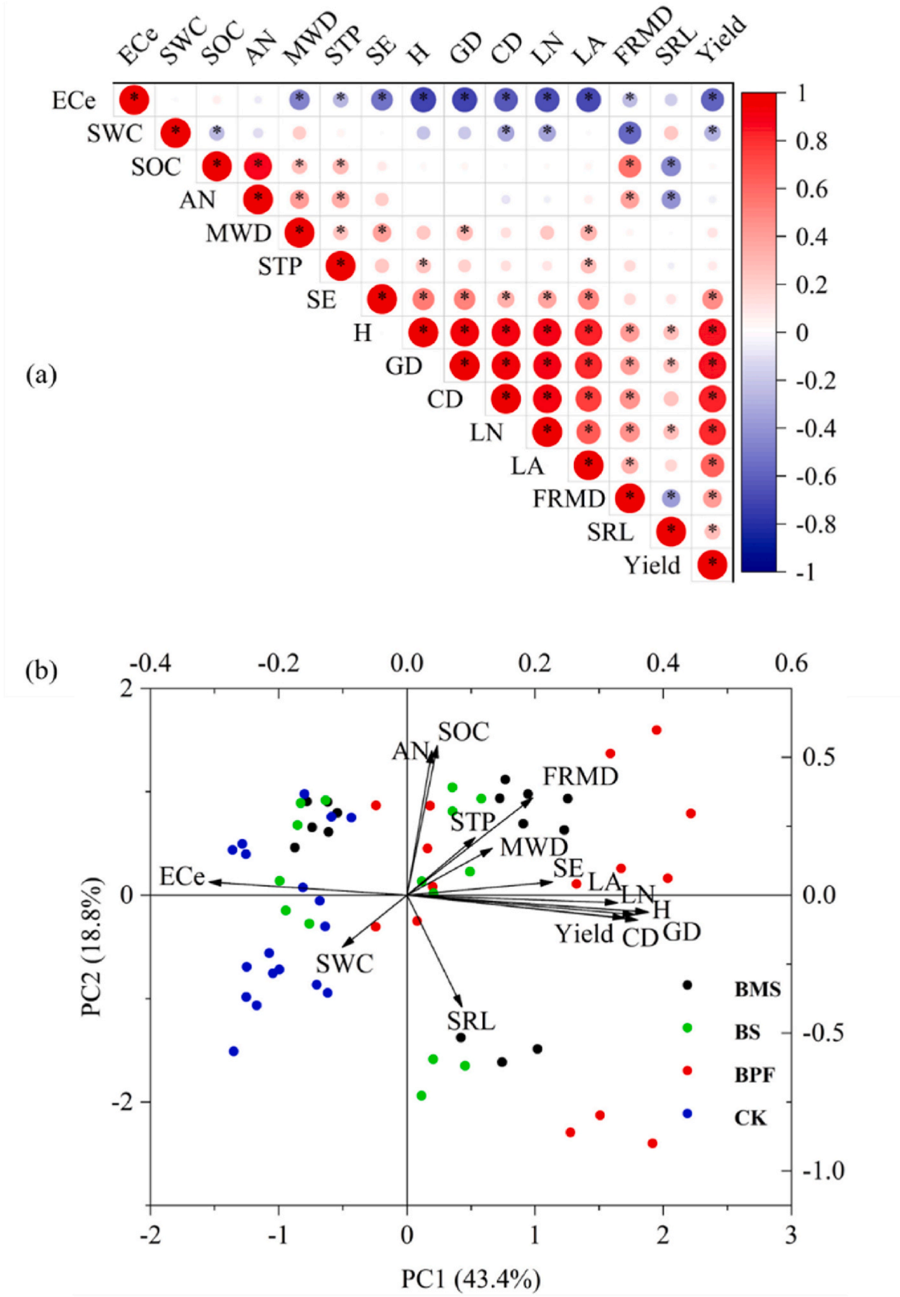


**Fig. 5.** (a) Relationships between plant growth indicators and soil properties across different capillary barriers and CK; (b) Coordination visualized by a principal component analysis (PCA).

their incomes, especially that of BPF, were greatly improved in the second year due to the yield improvements. As the saline soil environment improved, BPF took the lead in earning more net revenue after the two-year afforestation (Table 4).

## 5. Conclusion

This study proposed an effective desalinization approach in the root zone in the low-lying strongly salt-affected wasteland. Obviously, BPF better reduced the root-zone salinity than BMS and BS due to its greater effectiveness of salts isolation, and gravitational washing via more preferential pathways created by its more root penetration. BPF was superior in improving the topsoil fertility and structure, but inferior to BMS with extra organic straw in the subsoil. Plastic film as the material of capillary barrier was not only far less costly than that for BMS or BS, but also better improved plant establishment, growth, and yield, thus taking the lead in creating net revenue in the second year. However, given the nitrogen loss occurring for BPF, it's necessary for us to explore the rational application of absorbable nutrients (e.g., seaweed fertilizer) to alleviate the nutrient stress especially in early growth stage in future studies.

## CRediT authorship contribution statement

**Tao Liu:** Conceptualization, Investigation, Software, Writing – original draft. **Yuanbo Cao:** Conceptualization, Formal analysis. **Xuhu Wang:** Conceptualization, Formal analysis. **Qiqi Cao:** Investigation, Formal analysis. **Ruoshui Wang:** Methodology. **Yongmei Yi:** Formal analysis. **Yingtuan Zhang:** Formal analysis. **Huijie Xiao:** Funding acquisition, Supervision, Writing – review & editing. **Baitian Wang:**

**Table 4**
Comparative economic benefits among different capillary barriers.

| Treatments | Seed yields | Output values[a] | Input values ($/ha) | | | Net revenues |
|---|---|---|---|---|---|---|
| | (kg/ha) | ($/ha) | Material[b] | Labor[c] | Total | ($/ha) |
| *2016* | | | | | | |
| BMS | 0.64 | 2.32 | 117.07 | 583.02 | 700.09 | −697.77 |
| BS | 0.65 | 2.34 | 264.06 | 573.72 | 837.78 | −835.44 |
| BPF | 0.71 | 2.57 | 81.58 | 555.13 | 636.71 | −634.14 |
| CK | 0.57 | 2.06 | 75.30 | 415.68 | 490.98 | −488.92 |
| *2017* | | | | | | |
| BMS | 184.64 | 656.51 | 19.05 | 79.37 | 98.41 | 558.10 |
| BS | 177.67 | 631.70 | 19.05 | 79.37 | 98.41 | 533.29 |
| BPF | 393.47 | 1398.99 | 19.05 | 95.24 | 114.29 | 1284.70 |
| CK | 31.36 | 111.49 | 19.05 | 63.49 | 82.54 | 28.95 |
| *Sum* | | | | | | |
| BMS | 185.28 | 658.83 | 136.12 | 662.39 | 798.50 | −139.67 |
| BS | 178.32 | 634.04 | 283.11 | 653.09 | 936.20 | −302.16 |
| BPF | 394.18 | 1401.56 | 100.63 | 650.37 | 751.00 | 650.56 |
| CK | 31.93 | 113.55 | 94.35 | 479.18 | 573.53 | −459.97 |

[a] The average price of seeds was 3.61$ and 3.56$ per kg, respectively, in 2016 and 2017 (1$ = 6.64¥ in 2016 and 6.75¥ in 2017).

[b] Material input included seedling, pesticide, and buried material. About 26.67-$m^3$ maize straw (1 $m^3$ = 1.57$ in 2016), 26.67-$m^3$ sand (1 $m^3$ = 7.08$ in 2016), and 2.92-kg plastic film (1 kg = 1.87$ in 2016) were needed.

[c] Labor included the prices of rototiller operator, excavator operator, and other field assistants. The price of a rototiller operator for soil tillage and an excavator operator for drainage ditch was 30.12$ and 20.59$ per hr, respectively, in 2016. One labor per day for a field assistant cost 15.06$ and 14.81$ in 2016 and 2017, respectively.

Methodology, Conceptualization, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2022.133710.

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